

Computational Semantics of Claim Admissibility

Epistemic Debt, Typed Non-Answer States, and Governed Use under Declared Horizons

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August 2026

Abstract

Artificial intelligence can produce decision-ready semantic artefacts before their sources, transformations, limitations, permitted uses, and consequential effects are reconstructable. This paper defines a bounded computational semantics of claim admissibility and use-readiness. Given a versioned claim, evidence context, declared horizon, and resource policy, an effective evaluator returns evidence status, a componentwise Epistemic Debt vector, a typed terminal, and a receipt—not truth. A separate gate computes permission for one named use. Consequential action requires a current ActionCertificate bound to the exact action and control state, followed by an ExecutionReceipt. Classical computability is unchanged. Gödel, Church–Turing, Rice, and Shannon delimit four non-claims: complete self-grounding, a universal total decider, a universal checker of non-trivial program semantics, and information as warrant. The framework composes six functional contracts. Three concern query-relative scientific representation, bounded evaluation with honest non-answer terminals, and local epistemic transduction around probabilistic proposal and external certification. Three concern persistent observer composition and governance meta-control, componentwise residual-warrant accounting, and named-use/action mediation. Framework names follow their functions and carry no independent scientific standing. Local results address query factorisation, approximate and stochastic fidelity, budgeted terminalisation, and undecidability preservation or information obstruction. They also cover typed-terminal and candidate/certified preservation, conditional non-laundering, waiver without discharge, and relative layer removal. A bryophyte compilation and supplier-control path are stress tests, not validations. The programme loses when simpler systems reconstruct the same exports and achieve equal or better outcomes with lower burden.

Keywords: computational semantics; epistemic debt; claim admissibility; typed non-answer states; scientific representation; computability; abstention; provenance; assurance; bounded observers; action certificates; AI governance

Scientific status. Conceptual and formal-methodological. Gödel, Church–Turing, Rice, Shannon, factorisation, type-soundness, and finite-trace arguments are used only within their exact scopes; local propositions and conformance invariants remain conditional. Empirical superiority, Epistemic Debt coding reliability, dependency completeness, gate calibration, institutional legitimacy, and external validation remain open.

Standalone scope. Every functional contract is defined locally; framework names are introduced only as secondary labels. Expanded framework documents supply context and naming provenance, not independent evidence. No contract may borrow maturity from another, and every named layer has a relative-removal condition. Executable companions are treated as falsification surfaces, not independent evidence.

Reading key. Epistemic Horizon — versioned source, scope, query, transformation, receipt, validation, and conflict contract. **WF($\mathcal{H}$)** — internally evaluable. **adequate($\mathcal{H}, U, A$)** — sufficient for use U under authority A . **EvidenceStatus** — earned support. **Epistemic Debt** — componentwise unpaid warrant. **Typed Non-Answer States** — Unknown, Contradiction, OutOfHorizon, Review, Timeout, or Failure. **GateOutcome / ActionCertificate** — named-use permission and the current single-action token it may issue.

1. Claim admissibility: problem, thesis, and contribution boundary

A machine can produce a polished scientific synthesis, software diagnosis, legal issue list, compliance memorandum, or policy recommendation before an institution can reconstruct what licensed it. Truth remains indispensable. Use adds another question: what has the artefact earned under the sources, scope, transformations, checks, dependencies, and authority actually available? A claim may be true by accident, calibrated but unsupported, traceable but unauthorised, or executable under an illegitimate policy. These are distinct states with distinct remedies, not stylistic caveats.

The organising thesis is restricted. Claim admissibility and use-readiness can be computed locally when the object and contract are explicit. An epistemic computation is an effective procedure over a finitely encoded claim, declared horizon, and resource policy. It emits a typed epistemic state with a receipt. Its codomain is not merely Answer; it includes EvidenceStatus, Epistemic Debt, Terminal, and Receipt. Only a second, externally governed computation may return permission for a named use. A mature observer is not one that always answers; it terminates responsibly.

The contribution is architectural, not novelty by vocabulary. Epistemic logic and knowledge-based programs formalise what

agents know or believe and how knowledge tests affect distributed computation. Truth-maintenance, selective prediction, provenance, assurance, and runtime verification already cover adjacent functions (Hintikka, 1962; Doyle, 1979; Fagin et al., 1997; Chow, 1970; W3C, 2013; Bloomfield & Bishop, 2010). The integrated lifecycle does not replace them. Its narrower hypothesis is that typed interfaces can expose failures hidden when these registers are merged. Those interfaces connect representation, bounded evaluation, epistemic state, residual warrant, named-use authorisation, and action. The proposal loses distinctiveness whenever established methods reconstruct the same export with lower burden.

Use the six separations below as a reading key. They are type boundaries, not slogans. Observer relativity is not relativism: horizons can be compared through source manifests, queries, transmission contracts, validation rules, and authority relations.

Representation $\neq$ ontology A working vocabulary is not reality.	Computability $\neq$ validation A runnable evaluator may still be wrong.
Probability $\neq$ evidence Confidence does not create a source.	Evidence $\neq$ authorisation Support does not decide a use.
Execution $\neq$ legitimacy Following a rule does not legitimate it.	Visible workflow $\neq$ proof Logs still require integrity and interpretation.

1.1 Formal limits as negative architecture

Gödel, Church–Turing, Rice, and Shannon constrain the permissible claim space; none proves this architecture. Gödel applies to consistent, effectively axiomatized theories strong enough for arithmetic. Its local lesson is that internal derivability cannot become a complete truth or consistency certificate. Church and Turing delimit effective calculability and establish undecidable problems. A bounded observer therefore needs budgets, timeout, and honest partiality. Rice applies to non-trivial extensional properties of arbitrary computed partial functions; it does not make every finite, syntactic, or domain-restricted predicate undecidable. Shannon quantifies information and channel limits, not reference, justification, authority, or intended use (Gödel, 1931; Church, 1936; Turing, 1936/1937; Rice, 1953; Shannon, 1948). The constructive inference is local, not sceptical. Encode the question; declare the horizon; compute only effective or budgeted predicates; let non-answer be a typed success; preserve the path by receipt; keep permission external. Formal limitation becomes an engineering obligation. This also separates the proposal from classical epistemic logic. Modal formalisms reason about knowledge or belief; the present evaluator computes a claim’s institutional standing under source, transformation, validation, resource, and authority contracts. Epistemic logic may supply one local rule language inside the horizon, not the complete use-governance semantics (Hintikka, 1962; Fagin et al., 1995, 1997). Figure 1 marks the formal boundary; the positive contract chain follows the section order and is summarised in Table 1A.

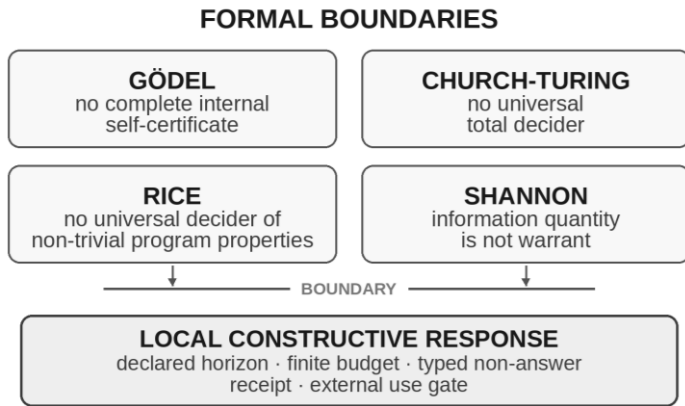


Figure 1. Reading rule: the upper row states four formal limits; the lower block states the local engineering response. Forbidden inference: the limits neither prove this architecture nor imply universal unknowability.

1.2 Claim classes

Every load-bearing statement belongs to one of six classes. STANDARD RESULT credits an established theorem instantiated locally. LOCAL PROPOSITION is derived inside the present definitions. CONFORMANCE INVARIANT holds only under its stated mediation and integrity premises. NON-ENTAILMENT marks what the architecture cannot infer. EMPIRICAL TARGET states a preregisterable comparison and its defeat condition. WITNESS shows constructibility, not validity or superiority. The labels implement Popperian loss conditions and prevent a diagram, prototype, or polished proof object from borrowing the authority of a theorem or experiment (Popper, 1959).

Residual contribution. The candidate delta separates evidence status, Epistemic Debt, terminal state, permission, and consequential action. It also tests horizon adequacy before permission, preserves reason-typed non-answer and candidate/certified states, binds permissive outcomes to one current action, and removes any named layer whose complete export is reconstructed more simply. Each item is defeated by a matched baseline that provides the same diagnostics and protection with lower burden.

2. Computational semantics of status, use, and action

2.1 The claim-admissibility object

Standard computation is not extended by fiat; its input and output types are. Once sources, claims, rules, policies, receipts, and budgets have finite encodings, an ordinary effective evaluator can compute a bounded status object. Computational semantics here names the subject matter and typed codomain of that evaluator, not a new model of computation. The first stage records what the claim has earned and what remains unresolved. The second decides what one use may do. Neither stage computes metaphysical truth.

Let the epistemic horizon be the typed record

$$\mathcal{H} = (S, \Omega, Q, \mathcal{T}, R_p, M, \text{Val}, \Pi),$$

Here S is the versioned set of admitted sources; Ω is the scope; Q the query families; and $\mathcal{T}$ the allowed transformations. R_p specifies required receipt types, M decidable support-and-contradiction rules, Val validation procedures, and Π conflict/priority policy. Scope and source set are different types; no relation such as $\Omega \subseteq S$ is assumed. The resource contract is a vector or policy, not merely a scalar.

Definition (well-formed horizon). $\text{WF}(\mathcal{H})$ requires typed, versioned fields. Where computation is claimed, transformations and validation procedures are effective on their declared domains. Receipt schemas match required and produced receipts. Invoked support/contradiction rules are decidable, and Π covers declared conflicts. $\text{WF}(\mathcal{H})$ says the contract can be evaluated, not that it is sufficient for a use.

Well-formedness is not use adequacy. Write $\text{adequate}(\mathcal{H}, U, A)$ when $\mathcal{H}$ satisfies the externally governed minimum-horizon contract for named use U under authority context A . $\text{WF}(\mathcal{H})$ does not entail $\text{adequate}(\mathcal{H}, U, A)$.

$$B = (b_{\text{time}}, b_{\text{steps}}, b_{\text{memory}}, b_{\text{cost}}, b_{\text{calls}}, \dots).$$

A semantic object v carries at least

$$x(v) = \langle \text{type}, \text{version}, \mathcal{H}, \text{sources}, \text{evidence}, \text{debt}, \text{authority}, \text{revocation} \rangle.$$

The local semantics of claim admissibility is

$$\text{CS}_{\mathcal{H}, B}(c) \rightarrow \text{EvidenceStatus} \times \text{DebtVector} \times \text{Terminal} \times \text{Receipt}.$$

DebtVector—hereafter Epistemic Debt—is the componentwise residual-warrant object; Section 7 develops its coding, propagation, discharge, and validation. A minimal evidence vocabulary distinguishes Uncomputed, ComputedUnsupported, Supported-

NotValidated, Validated, Contradictory, and OutOfScope. The terminal sum is defined below.

$$\text{Term}(Y) = \text{Answer}(Y) + \text{Unknown} + \text{Contradiction} + \text{OutOfHorizon} + \text{Review} + \text{Timeout} + \text{Failure}.$$

The semantic evaluator is partial unless each invoked operation is effective. Under a budgeted dispatcher it is totalised into $\text{Term}(Y)$, not into $\text{Answer}(Y)$. A responsible timeout or contradiction is therefore a successful typed computation.

Permission is a second computation. For named use U , governance context C_g , and an externally governed minimum-horizon contract $\alpha(U)$,

$$\text{Gate}_{U,C_g}(\mathcal{H}, \text{CS}_{\mathcal{H},B}(c), \alpha(U)) \rightarrow \text{GateOutcome}.$$

with $\text{GateOutcome} \in \{\text{Warranted}, \text{PermittedWithWaiver}, \text{ReviewRequired}, \text{HorizonInadequate}, \text{Blocked}\}$ and a GateReceipt . The gate first tests $\text{adequate}(\mathcal{H}, U, A)$. A well-formed but narrow horizon may still be unfit for the requested use. EvidenceStatus records what evidence and checks earned. Epistemic Debt records what remains unpaid. GateOutcome records what one use may do. Confidence is only a model-side estimate; it is neither evidence status nor permission. Figure 2a isolates claim-state computation from named-use permission and exposes the certificate boundary.

The epistemic head is only a rendering and routing boundary:

$$\text{HeadOutput} = \text{render of } \langle \text{EvidenceStatus}, \text{DebtVector}, \text{Terminal}, \text{optional GateOutcome} \rangle.$$

It presents or invokes the relevant computations; it does not certify its own inputs, decide truth, or manufacture authority.

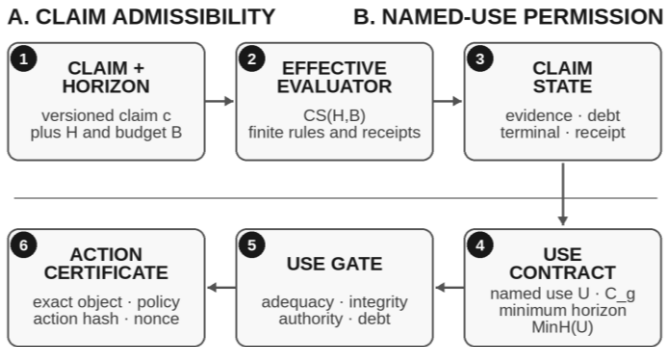


Figure 2a. Reading rule: ClaimState is computed before and independently of permission for a named use; only a permissive outcome may issue a current ActionCertificate. Forbidden inference: evidence status or debt alone authorises action.

2.2 Action certificates, canonical export, and invariants

For a consequential action a , a permissive gate outcome is not yet executable. An ActionCertificate is an abstract, single-action authorisation token. It binds permission to one exact action and one current control state; it is not proof, truth, or legitimacy. Its integrity/authenticity binding is implementation-specific—for example, a digital signature, MAC, protected local state, transaction identity, or trusted channel:

$$\text{Cert}_U = \langle \text{claim_id}, \text{object_ver}, \mathcal{H}_{id}, \text{policy_id}, \text{use_id}, \text{action_hash}, \text{authority_id}, \text{resource_scope}, \text{expiry}, \text{nonce}, \text{GateOutcome} \rangle.$$

Execution is a separate bounded transition:

$$\text{Exec}(a, s, \text{Cert}_U) \rightarrow (s', \text{Receipt}_{\text{exec}}) + \text{Review} + \text{OutOfHorizon} + \text{Failure}.$$

Exec is admissible only when $\text{hash}(a)$ and all load-bearing object, horizon, policy, authority, credential, scope, expiry, nonce, and resource fields still match Cert_U . Gate evaluation and dispatch must be atomic with respect to those fields. Otherwise a commit-time check must establish that none changed. Any mismatch returns Review or Failure; a previous certificate cannot be replayed against a new state. This is ordinary systems engineering coupled explicitly to evidence, debt, and named use.

C. EFFECT, RECEIPT, AND REVALIDATION

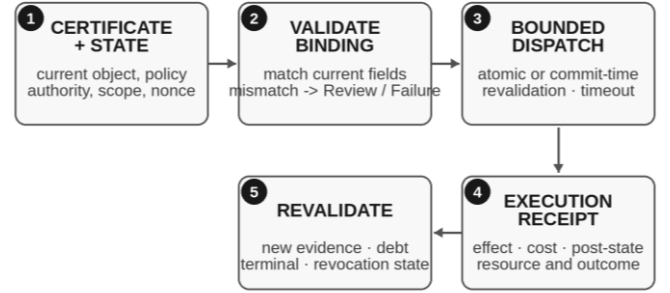


Figure 2b. Reading rule: current-state binding is checked before effect; bounded dispatch then emits an execution receipt and triggers post-state revalidation. Forbidden inference: successful execution, or a stale certificate, establishes legitimacy.

Every layer exports a common audit envelope:

$$\text{Export}_B = \langle \text{value_or_terminal}, \rho, \Delta E, \Delta d, \text{policy/version, threat/authority metadata, action_binding, resource_cost}, Q\text{-observables} \rangle.$$

Here ρ is the receipt, ΔE the evidence-state effect, and Δd the debt effect. For non-consequential outputs, action_binding is absent. When action is possible, it records the certificate/dispatch relation. This envelope supports composition and relative layer removal. A named layer is dispensable when adjacent components reconstruct its entire export for the declared query family without weakening terminals, receipts, debt, action binding, threat/authority metadata, resource accounting, or gate outcomes. Conformance is tested family by family. The matrix below states the premise each family requires and one negative test that should make failure visible. These are not global correctness guarantees.

Invariant family	Required premise → negative test
Boundedness	Current $\mathcal{H}$ and B identify the run → remove/change either and require refusal or re-evaluation.
Query relativity	Q is declared → find one fibre with equal image but different q-values.
Terminal completeness	Every unresolved condition maps to Term → delete a mapping and test whether Answer is fabricated.
Candidate/certified separation	Certification needs external authority and receipt → relabel Candidate as Certified.
Receipt continuity	State changes expose inputs, policy, effects, and failure → remove one required receipt field or readback.
Debt conservation	Relevant dependencies and Join Pd are complete enough → seed a hidden edge or unreceipted debt drop.
Authority separation	Authority, scope, freshness, and revocation are current → let executor self-certify or reuse stale authority.
Pre-action mediation	Current Cert U is bound to dispatch → bypass, substitute, replay, or duplicate the action.
Version invalidation	Load-bearing versions are tracked → mutate source, model, policy, or horizon after certification.

2.3 One complete path: a supplier-control claim

Worked specification, not outcome evidence. Let c be “payments above the threshold require CFO approval.” The path below uses illustrative candidate debt codes to show where one claim changes state.

Stage	Supplier-control state and visible loss condition
1 · Horizon / sources	$\mathcal{H}_{\text{narrow}}$ admits two interviews. $\mathcal{H}_{\text{assurance}}$ also requires written policy, exception procedure, transaction sample, and independent reviewer. Both may be WF; only the stronger horizon may be adequate for formal assurance.
2 · ClaimState	Under $\mathcal{H}_{\text{narrow}}$: SupportedNotValidated + Answer(c) + receipt p_1 . The receipt links both interviews and the extraction transform.
3 · Candidate debt	Illustrative only: $d=(0,0,1,1,2,0,2,?,2)$ for source, trace, transformation, collapse, label, reification, validation, authority, horizon/scale. “?” is unassessed, never clean.
4 · Named-use gate	Drafting use may be ReviewRequired or Permitted-WithWaiver. Formal control assurance is HorizonInadequate until its minimum contract is met.
5 · Certificate	No certificate is issued for an automatic policy update while the gate is non-permissive. After discharge and Warranted, Cert_U binds the exact action and current versions.
6 · Commit / execution	Any changed object, policy, authority, credential, action hash, expiry, or nonce returns Review/Failure. Bounded dispatch emits Receipt_exec and post-state revalidation.
7 · Simpler baseline	Endpoint-matched baseline: provenance + assurance for reconstruction; add selective prediction for non-answer and firewall or transactional control for action. It wins if every tested state and refusal is reconstructed at lower burden.

Under $\mathcal{H}_{\text{narrow}}$, the claim can be SupportedNotValidated while the horizon remains inadequate for formal assurance. An internally correct evaluation can therefore refuse permission because the contract is too weak for the named use. The strip below projects that lifecycle across domains; it is illustrative, not evidence.

Context	Epistemic state capsule	Boundary
Scientific synthesis	Debt: transformation / validation / horizon may be open. Gate: ReviewRequired; HorizonInadequate if the stronger named use fails adequacy.	No numeric d.
Clinical decision support	Debt: validation / authority / horizon open; transformation may be open. Gate: ReviewRequired or HorizonInadequate for diagnosis/treatment.	Hypothetical; no clinical authority.
Low-risk e-commerce	Debt: fit / validation may be open. Gate: Warranted only if adequacy and policy hold; display-only use needs no action certificate.	Low risk is not $d=0$.

Coding guardrail. These are qualitative profiles, not numeric d ; coding requires a declared record, and absent debt fields mean not represented/not assessed, never $d=0$. **Integration discriminator.** Match each baseline to its endpoint; remove the added layer if the matched composition reconstructs the same state and refusal at lower burden.

3. Scientific representation across levels

Scientific practice rarely represents a domain through one complete description. Models idealise, select variables, mediate between theory and experiment, and coexist for different purposes; in biology, no causal level is privileged in advance (Noble, 2012). Model-based and perspectival accounts make adequacy purpose- and question-relative. Blackwell comparison and abstract interpretation provide rigorous ancestors for task-relative informativeness and sound abstraction (van Fraassen, 1980; Giere, 1988; Blackwell, 1953; Cousot & Cousot, 1977; Weisberg, 2013; Frigg & Nguyen, 2020). The representation contract used here—Reticular Local Abstraction (RLA)—takes no position on final ontology. It types scientific levels and requires even-

ry cross-level inference to declare what it preserves, what it loses, and for which queries.

A level is

$$L_i = \langle D_i, \Sigma_i, Q_i \rangle,$$

where D_i is a typed state domain, Σ_i the admitted operations or dynamics, and Q_i the admissible queries. A transmission $\tau_{ij}: D_i \rightarrow D_j$ is a partial or stochastic scientific operation, not an ontological bridge. Each edge declares domain, codomain, query family, side information, tolerance, randomness, receipt, rollback, and open debt. The residual content of the contract is therefore not that levels exist; it is that representational adequacy, loss, and transfer are made inspectable at their interfaces.

STANDARD RESULT 1 — query factorisation. For deterministic $\tau: X \rightarrow Y$ and $q: X \rightarrow A$, exact preservation of q through τ holds iff q is constant on every fibre of τ . Equivalently, there exists $\bar{q}: \text{im}(\tau) \rightarrow A$ such that $q = \bar{q} \circ \tau$. Extending $\bar{q}$ to all of Y requires an additional assumption and is not part of the result. Full state recovery is the special case where Q separates all source states. Thus injectivity is sufficient for every point-separating query, but far stronger than query-relative fidelity.

Proof sketch. Define $\bar{q}(\tau(x)) = q(x)$ on $\text{im}(\tau)$. Constancy of q on every fibre makes $\bar{q}$ well-defined; conversely, $q = \bar{q} \circ \tau$ immediately implies fibre constancy. The result is ordinary factorisation, not a new information theorem.

Approximate fidelity must name a metric and tolerance. For q with codomain metric d_q ,

$$\Delta_q(\tau) = \sup_{y \in \text{im}(\tau)} \sup_{x, x' \in \tau^{-1}(y)} d_q(q(x), q(x')) \leq \varepsilon.$$

For a stochastic transmission $P_\tau: X \rightsquigarrow Y$, recovery requires a decoder kernel $D: Y \rightsquigarrow A_q$, loss ℓ , input law μ , and tolerance ε , for example

$$\mathbb{E}_{x \sim \mu, y \sim P_\tau(\cdot|x), a \sim D(\cdot|y)} [\ell(a, q(x))] \leq \varepsilon.$$

A recorded seed can replay one realised run; it does not establish distributional validity. A material change in the input law activates version invalidation. Shannon information and information-bottleneck objectives quantify uncertainty or compression, not evidence, authority, or use (Shannon, 1948; Tishby, Pereira, & Bialek, 1999). Causal abstraction is stronger whenever interventions and causal semantics belong to Q , especially under lossy, mechanistic, compositional, distributionally robust, or cross-population transport settings. Current work provides component-level abstractions and approximate robustness certificates. Model-level transport maps can also bound target queries when exact transport fails (Beckers & Halpern, 2019; Beckers, Eberhardt, & Halpern, 2020; Geiger et al., 2025; Xia & Bareinboim, 2025; Felekis et al., 2025, 2026; Lorenz & Tull, 2026). In that regime RLA survives only as the lifecycle contract coupling query fidelity to receipts, residual warrant, and named use. It is removed if the causal formalism supplies that export more simply.

Non-injectivity alone does not establish emergence. A constant map is maximally many-to-one yet creates no scientifically significant macro-property by itself. A defensible emergence claim must declare micro- and macro-languages, admissible algorithms, resource bounds, encodings, and invariance under reasonable recodings (Anderson, 1972; Shalizi & Crutchfield,

2001). No later result in this paper depends on an emergence theorem. Collapse matters here because it blocks queries, changes reconstruction cost, and can generate debt when lost distinctions are later required.

4. Executable bounded evaluation

Compact Reticular Computability (CRC) is not a computability class; it is an acceptance contract for an executable bounded reticulum. The representation contract states what a scientific model preserves, loses, and for which questions. CRC asks a different question: has that typed reticulum been compiled into a standard effective transition system that can finish honestly? It requires an effective presentation of typed states and operators, a scheduler, a resource policy, a terminal sum type, receipts, and bounded adapters for external operations. “Compact” means operationally specified and auditable, not finite reality, complete science, or metaphysical closure. Define

$$\text{CRC} = \langle \mathcal{R}, \Gamma, \sigma, B, \text{Term}, \mathcal{H} \rangle,$$

where $\mathcal{R}$ is the typed reticulum, Γ the configuration space, σ an effective scheduler, B the resource policy, Term the terminal sum, and $\mathcal{H}$ the horizon. A configuration includes the current node, typed local value, evidence and governance state, immutable ledger prefix, policy identifier, and remaining resources.

CRC separates three obligations that are often conflated. Local effectivity: every operator and edge is computable on its declared finite encoding or returns a terminal. Budgeted totalisation: a dispatcher tracks a rank $v:\Gamma \times B \rightarrow W$ into a well-founded set W ; every nonterminal step strictly decreases v or consumes finite budget, and operational faults map to Timeout or Failure. External calls must be cancellable or time-bounded, or must return control to a timeout-capable asynchronous dispatcher. Global termination: an acyclic graph with terminating nodes suffices; a cyclic network additionally needs a well-founded rank, bounded iteration, fixed-point criterion, or explicit timeout. Local termination does not compose automatically: two individually terminating units can alternately re-enable one another forever.

LOCAL PROPOSITION 2 — budgeted totalisation. Assume effective local transitions and scheduling. Every nonterminal dispatch must strictly decrease a well-founded rank or consume finite budget. Every external operation must use a bounded/cancellable adapter or timeout-capable dispatcher, and every operational exception must map into Term . Under these premises, the wrapped evaluator terminates with exactly one typed terminal for every finite admissible input. Proof sketch: no infinite strictly descending chain exists in W ; finite budget exhaustion maps to Timeout; adapter and operational faults map to Failure. Without progress, interruption, or timeout, the evaluator remains partial (Church, 1936; Turing, 1936/1937; Plotkin, 2004; Baier & Katoen, 2008; Leucker & Schallhart, 2009).

Gödel, Turing, and Rice now do different work. Gödel blocks promotion of internal proof into complete self-certification. Turing requires the evaluator to expose partiality, nontermination, and budgets. Rice forbids a universal decider for non-trivial extensional program properties but leaves syntactic checks, finite domains, and explicitly restricted predicates available. CRC is thus an operational contract for where ordinary computation may answer and where it must return a typed boundary.

Type soundness divides into preservation and progress-or-terminal. Edge well-typedness preserves the configuration type across a transition; terminal completeness ensures that a well-typed nonfinal configuration either advances or returns a de-

clared terminal. This is continuous with operational type-soundness and model-checking methods, not a new theorem about arbitrary workflows (Wright & Felleisen, 1994; Clarke, Grumberg, & Peled, 1999).

STANDARD REDUCTION / OBSTRUCTION 3. Let p be an undecidable predicate on X and $\tau:X \rightarrow Y$ computable. If $p = \bar{p} \circ \tau$, then $\bar{p}$ cannot be decidable, or p would be decided by composition. Conversely, if $\tau(x) = \tau(x')$ while $p(x) \neq p(x')$, no target-only decoder can answer p correctly on both states. A collapse can therefore block a hard query; it does not solve it. Rice’s theorem constrains nontrivial extensional properties of arbitrary programs, not every bounded predicate in a domain-specific CRC (Rice, 1953).

4.1 The bryophyte model as a compilation and falsification stress test

A generalist bryophyte model is used only as a compilation stress case. It is chosen because the domain combines heterogeneous sources, cross-taxon and cross-scale transfer, feedback cycles, calibration gaps, and observable-family tests—not because bryology is central to the general theory. Bryophytes link desiccation tolerance, photosynthetic recovery, cellular stress, tissue and colony structure, substrate, microclimate, reproduction, and biotic interaction across heterogeneous literatures (Proctor, 2001; Oliver, Velten, & Mishler, 2005; Glime, 2007). A literature-derived reticulum can encode twenty abstraction levels, forty-two parameters, update rules, thresholds, scenarios, and a finite execution loop. This witnesses constructibility; it does not show that the plant has been reproduced or that the selected literature is complete.

Scientific relations rarely arrive as interoperable code. A needed relation may be directly measured, transferred from a neighbouring taxon, reported only as a range, synthesised from several sources, approximated by the compiler, or unresolved. These are orthogonal dimensions rather than one status:

$$\begin{aligned} \text{RelationRecord} = & \text{source_basis} \times \text{compiler_ops} \\ & \times \text{calibration_state} \times \text{target_use} \times \text{locator} \\ & \times \text{conditions} \times \text{observable_family} \times \text{discharge_test}. \end{aligned}$$

$\text{source_basis} \in \{\text{Direct}, \text{Transferred}, \text{Synthesised}, \text{Unresolved}\}$; compiler_ops is a set that may contain *Approximated*; calibration_state records what was actually fitted or tested; target_use declares what the relation may currently support. “Direct” does not imply validated; “transferred” may also be approximated. The record activates component-specific debt but does not assign severity automatically.

For example, a low-surface-water $\rightarrow$ ABA relation can be *Direct* or *Transferred*, independently carry $\text{compiler_ops} = \{\text{Approximated}\}$, and remain *Uncalibrated* for the target microclimate. A smooth executable curve therefore does not pay transfer or calibration debt; it only makes those obligations testable.

EMPIRICAL TARGET — observable-family discrimination. Synthetic and empirical trajectories may be compared under preregistered observables, sampling conditions, thresholds, expert tasks, and trivial baselines. Failure to discriminate supports adequacy only for that observable family. It does not establish biological equivalence, total closure, or a privileged ontology. Conversely, failure defeats the declared implementation or observable-family adequacy claim, not every complex-system application of the framework. The model is valuable because it turns missing science into addressable calibration and validation obligations.

Executable falsification surface. The ROA Bryophyte Epistemic Lab v9 compiles Annex B into a standalone browser runtime with twenty levels, forty-two parameters, seeded replay, interventions, halting, trace export, and provenance labels. SOURCE, MODEL_DEFAULT, REPAIR, and DEBT_OPEN remain distinct. Domain experts can challenge the implementation through blind real-versus-synthetic discrimination, climate and recovery tests, parameter mutation, reticular-closure tests, trace removal, horizon transfer, and matched simpler baselines. Passing internal checks establishes constructibility and reconstructibility only; biological adequacy remains an external empirical obligation.

The optional PCE/cellular-automata bridge is outside the causal spine. A cross-formalism equivalence claim requires a bounded simulation certificate naming source and target systems, encoding, observables, distortion, overhead, and trace horizon. It must also state reverse obligations when equivalence, rather than one-way simulation, is claimed. Visual or dynamical resemblance alone carries no standing.

5. Local epistemic transduction

Local epistemic transduction is the smallest governed operation in the lifecycle. Neural implementation is optional, and candidate generation may be stochastic; the surrounding control contract must remain effective and auditable. An Epistemic Computational Unit (ECU) does not “understand” by declaration. It applies effective transformations and policy checks to finite encodings of claims, evidence, context, and resources under a versioned epistemic matrix

$$M_u = (E_u, R_u, C_u, P_u),$$

where E_u names domain entities, R_u inference rules, C_u coherence/contradiction constraints, and P_u abstention, rejection, escalation, and fallback policies. M_u is a local rule-and-policy object; it is not organisational authority. The unit then emits a typed value, evidence/debt effects, a receipt, or a non-answer terminal:

$$u: X \times \mathcal{H}_u \times \mathcal{E} \times C_g \times B \rightarrow (Y \times \rho \times \Delta\mathcal{E} \times \Delta d) + \text{Term}(Y).$$

Each ECU declares an effect mask $\text{Eff}_u = (\text{Read}_u, \text{Write}_u)$. The receipt lists the fields actually read and changed. A unit cannot silently modify certification, authority, policy, revocation, or an unrelated debt component merely because its output schema contains those fields. This makes epistemic state change an auditable effect rather than prose surrounding a model call.

A generic local semantic aggregation makes the “convolution” analogy operational:

$$h_{k+1}(v) = \text{Agg}_k \langle u_{k,m}(h_k(w), M_{k,m}) : w \in N_k(v), m \in I_k \rangle :$$

Neighbourhood $N_k(v)$, local operators $u_{\{k,m\}}$, and aggregation Agg_k are declared and typed. They may be neural, symbolic, database, human-review, or mixed operators; the equation is a semantic local-operator schema, not a claim that ECNN requires CNN layers.

Historically termed an Epistemic Convolutional Neural Network (ECNN), the typed reticulum preserves only a non-constitutive CNN analogy. Local receptive fields become local semantic operators; pooling becomes declared collapse; feature stabilisation becomes candidate reification (LeCun et al., 1998; Goodfellow, Bengio, & Courville, 2016). The reticulum contains ECUs,

representation operators, a scheduler, an epistemic head, and a visible state register. It may use neural models, symbolic rules, databases, humans, or mixed services. Its distinctive claim is a common epistemic interface, not a new neural primitive.

$$\mathcal{N} = \langle V_N, E_N, \lambda, \kappa, \sigma, \text{Pol} \rangle,$$

where V_N is the versioned semantic-object set and E_N the typed edges. λ assigns abstraction level. κ assigns epistemic state: candidate_AI, certified_human, disputed, superseded, unknown, or contradiction. σ is the scheduler; Pol is the policy/validation bundle.

The architecture has two planes. A probabilistic proposal plane may retrieve, classify, summarise, induce a label, or propose a relation. Every stochastic output is persisted as a candidate event with input, model, configuration, and output hashes. A deterministic control plane checks types, permitted transformations, query fidelity, contradictions, receipts, policy, and gate conditions. Deterministic replay is claimed only for the control plane when sources, operators, scheduler, budgets, policies, and recorded exogenous proposals are fixed. Temperature zero or a named checkpoint is not proof of bit-level reproducibility.

$$\text{Candidate}(v, \rho_p) \xrightarrow{\text{Certify}_{a,\pi,\rho_c}} \text{Certified}(v, \rho_p, \rho_c).$$

Certification is an externally authorised, policy-governed, receipted state transition. The actor or validator a , policy π , and certification receipt ρ_c are recorded; schema validity, fluency, model confidence, or repeated agreement cannot self-certify a candidate.

The epistemic head renders EvidenceStatus, DebtVector, Terminal, and—when a use is named—GateOutcome. It may invoke the gate but does not form a truth judgement or self-certify. This places ECU/ECNN beside partial transducers, many-valued or epistemic logic, truth-maintenance, and knowledge-based programs. It also overlaps neural-symbolic reasoning, selective prediction, calibration, conformal prediction, semantic uncertainty, provenance, and human review (Belnap, 1977; Doyle, 1979; Fagin et al., 1997; d’Avila Garcez, Gabbay, & Lamb, 2012; Chow, 1970; Guo et al., 2017; Vovk, Gammernan, & Shafer, 2005; Farquhar et al., 2024; Wen et al., 2025). Those methods can inhabit or replace the layer. The residual interface is reason-typed non-answer, receipted effects, candidate/certified separation, and downstream preservation.

CONFORMANCE INVARIANT 4 — typed non-answer and certification preservation. In a well-typed ECU network, no edge or scheduler rule may coerce Unknown, Contradiction, OutOfHorizon, Review, Timeout, or Failure into Answer. Nor may it promote Candidate to Certified without an explicit state-changing event that supplies the missing evidence, resolution, authority, or policy transition and emits a receipt. Type preservation is machine-checkable; evidential adequacy still depends on complete enough dependencies and valid external inputs. Delete or relabel a non-answer or candidate on one edge: if an authoritative answer remains possible, the network has an epistemic type error.

The distinctive ECNN claim is therefore not abstention alone. It is reason-typed non-answer plus a receipt, downstream preservation, candidate/certified separation, and use-sensitive routing. The layer is empirically unnecessary if a simpler reject option or workflow provides equal diagnosis, reconstruction, and protection with lower burden.

6. Observer compilation and meta-control

Persistent observer composition turns local contracts into a versioned, reconstructable system. The resulting observer contract—called a Reticular Observer Architecture (ROA)—is not merely a graph. Observer compilation is the map

$$\text{Compile}(\text{Field}, \mathcal{H}, \text{Policy}) \rightarrow O + \text{ObserverManifest},$$

where

$$O = \langle \mathcal{H}, \mathcal{R}, \mathcal{N}, \text{Mem}, \text{Mon}, \text{Threat}, \text{Gov}, \text{Auth}, B \rangle.$$

The manifest includes a type environment, source manifest, invariant registry, claim-to-evidence map, terminal and failure policy, authority map, monitoring plan, action-surface inventory, and rollback/revocation plan. Given a query, O returns a reconstructable path, terminal, receipt, and effects rather than a free-standing string. The answer is the path: persistent intermediate artefacts - definitions, atoms, candidate relations, contradictions, reviews, waivers, and revocations - make cumulative intelligence reside in the governed substrate and accountable human decisions. Generative models remain replaceable proposers and navigators; they do not own the knowledge or certification state. A candidate becomes reusable only through a receipted reification transition; that transition is where representational loss, status promotion, and downstream warrant become jointly addressable.

Observer compilation is maintained by semantic CI/CD. Source manifests, schemas, query contracts, candidate labels, dependency edges, validators, and authority roles are versioned and diffed. Action surfaces and gate thresholds are versioned too; negative tests challenge changes, with rollback or revocation paths retained. Intermediate artefacts remain first-class. The proof chain is source $\rightarrow$ transformation $\rightarrow$ receipt/readback $\rightarrow$ evidence status $\rightarrow$ witness/review $\rightarrow$ named-use decision $\rightarrow$ action certificate $\rightarrow$ execution receipt. Correctness is checked at the relevant boundary, and an artefact may travel with a checkable certificate; receipts are not thereby formal proofs (Saltzer, Reed, & Clark, 1984; Nacula, 1997). Output is not proof, and confidence is not evidence. A log is not automatically a receipt; export is not witness; review is not approval. Model memory is not a source of truth. Anti-false-green tests remove a source, receipt, reviewer, dependency, policy version, or certificate field and require authoritative success or dispatch to disappear.

ROA is constrained by provenance, runtime verification, assurance cases, proof-carrying execution, and end-to-end system principles. Provenance models entities, activities, agents, and derivations. Runtime verification checks declared properties of observed runs. Assurance cases organise claims, arguments, evidence, and defeaters. Algorithmic audits make lifecycle accountability explicit (Buneman, Khanna, & Tan, 2001; Cheney, Chiticariu, & Tan, 2009; Leucker & Schallhart, 2009; Bloomfield & Bishop, 2010; Raji et al., 2020; Paterson et al., 2025; Saltzer, Reed, & Clark, 1984; Nacula, 1997). ROA adds no warrant by naming these ingredients. Its candidate delta is the coupled export of typed terminals, evidence effects, residual debt, horizon adequacy, authority, revocation, action binding, and named-use permission. Artificial epistemic agents, execution provenance, agent trails, and certified traces are comparator families, not supporting evidence (Marchal et al., 2026; Liu Yanglet, Wang, & Capponi, 2026; Wang et al., 2026; Wu et al., 2026). If those systems and composable assurance reconstruct Export_B with lower burden, ROA should be compressed or removed (Zhao, 2026).

Monitoring never establishes completeness. Non-laundering requires complete mediation of authoritative-use and consequential-action paths, trace integrity, current versions, visible

revocation, sufficient dependency coverage, role separation, and detectable bypasses. NIST documents fragmented logs, human burden, drift, and immature monitoring methods as practical threats to those premises (Rao et al., 2026). The governing maxim is symmetrical: the horizon protects computation from inflation, while the named-use/action gate protects the institution from a self-serving horizon. A perfect internal trace with one unmonitored export or tool surface is still an incomplete observer.

6.1 Pre-action mediation and bounded execution

For an acting system, a consequential tool call is itself a named-use transition. ROA must interpose before effect: the gate issues an ActionCertificate only for a specific action hash and current object, horizon, policy, authority, credential, scope, and resource versions. Gate evaluation and dispatch are atomic with respect to those fields, or they are revalidated at commit. Approve-then-mutate, stale-certificate replay, duplicate non-idempotent dispatch, or changed credentials therefore refuse or reopen review. Hanging tools must be cancellable/time-bounded or return control to a timeout-capable dispatcher. Partial multi-step effects require declared idempotency, rollback, or compensation semantics. AEGIS, certified traces, and agentic-transaction systems are direct substitution baselines for this action-control surface (Yuan, Su, & Zhao, 2026; Liu Yanglet, Wang, & Capponi, 2026; Sun, Wang, & Li, 2026).

6.2 Governance meta-control

At the governance boundary, the governance meta-controller is a computational controller, not a moral agent, truth oracle, or self-authorising authority. Its state is

$$z_t = (W_t, S_t, K_t, \text{Hist}_t),$$

where W is the world/evidence model, S the operational self-model of limits and reliability, K the versioned normative kernel, and Hist the epistemic history. New evidence e_t and an authorised intervention a_t update the state:

$$T(z_t, e_t, a_t) \rightarrow z_{t+1}.$$

Case synthesis and system-level escalation are distinct:

$$\begin{aligned} G_{\text{case}}(z_t, q_t) &\rightarrow \text{CaseArtifact}, \\ G_{\text{crit}}(z_t, \text{trace}_t) &\rightarrow \text{EscalationArtifact}. \end{aligned}$$

A critical output carries an EscalationReceipt(owner, reason, evidence_gap, required_authority, expiry_or_next_action). Escalation may interpose before effect, not only report after completion. The human decision is an external, versioned, receipted state-changing event whose authority and scope are recorded back into Hist ; it is not another model inference. The controller monitors coverage, conflicts, resource state, drift, policy triggers, unresolved debt, queued effects, and stop requests. “Meta-cognitive” is used only in this engineering sense. Its roots are resource-bounded metareasoning, autonomic control, runtime monitoring, and evidence that active human participation can surface errors missed by post-hoc review (Russell & Wefald, 1991; Kephart & Chess, 2003; Leucker & Schallhart, 2009; Sabouri et al., 2026).

NON-ENTAILMENT 1 — execution does not establish legitimacy. Let K_1 and K_2 be internally consistent normative kernels with incompatible prescriptions for the same case. A conforming evaluator can prove that a run followed K_1 or K_2 . Execution alone cannot choose which kernel is legitimate without external authority, contestation, effective-date, and revocation relations. The governance meta-controller can expose the boundary and route escalation; it cannot authorise itself or decide moral truth.

7. Residual warrant and use gating

The preceding stages represent, evaluate, certify, and compose semantic objects. Epistemic Debt is the cross-cutting accounting remainder that survives those transformations until typed discharge or a bounded, visible waiver. The phrase predates this framework and has been used for technical ignorance in smart manufacturing; debt metaphors also inherit from software and machine-learning engineering (Ionescu, Schlund, & Schmidbauer, 2019; Cunningham, 1992; Sculley et al., 2015). The moat is not the name. It is the typed separation among EvidenceStatus, componentwise unpaid warrant, GateOutcome, and waiver. The construct is an accounting layer for the bounded semantics, not a probability of falsehood or a moral score.

Controlled reification is the transition that makes debt attachable. A pattern, sentence, relation, or label becomes manipulable only after it receives stable identity and construction metadata. The record names its source pattern, transformation operator, lost distinctions, evidence state, allowed and refused uses, owner, and rollback or discharge route. Reification is not an error: cumulative science and institutional work require reusable objects. The failure is silent status promotion—an AI candidate treated as a validated control, a transferred equation treated as locally established, or a summary treated as the underlying evidence. The object may exist before it is warranted, but it must carry the conditions of its construction. This is the local bridge from semantic computation to the accounting layer.

$$\mathbf{d} = (d_s, d_t, d_x, d_c, d_l, d_r, d_v, d_a, d_h) \in \{0, 1, 2, 3, ?\}^9.$$

The vector covers source, trace, transformation, collapse, label, reification, validation, authority, and horizon-or-scale debt. The symbol ? means unassessed, not infinite or zero. The vector is a candidate ordinal/categorical coding instrument, not an interval scale, semantic-entropy estimate, or probability of error. Semantic uncertainty is distinct even when its detection is useful (Farquhar et al., 2024). Each component has its own diagnostic and discharge path; summing them can destroy that path.

Minimal coding contract. 0 means the obligation does not arise under the horizon. 1 marks a documented bounded caveat; 2 a material unresolved obligation; 3 a critical unresolved structural obligation. ? means unassessed because coverage is insufficient. The codes are use-independent obligation states, not gate verdicts or additive scores. They remain candidate anchors pending a coding manual, training set, adjudication rules, and independent componentwise reliability.

Debt component	Diagnostic question → typical discharge
Source	Required sources admitted, current, and located? → attach/verify or narrow scope.
Trace	Derivation and replay reconstructable? → repair receipt, provenance, and readback.
Transformation	Operations explicit and fit for use? → verify operator, version, approximation, and effects.
Collapse	Which merged distinctions are later needed? → preserve side information/decoder or restrict use.
Label	Does the name overstate evidence, scope, or modality? → relabel.
Reification	Candidate promoted without a receipted transition? → record identity, owner, provenance, rollback/certification.

Debt component	Diagnostic question → typical discharge
Validation	Did the declared test run and pass? → execute use-specific validation with a witness.
Authority	Reviewer/issuer competent, current, and separated? → authorised review plus revocation state.
Horizon / scale	Does support transfer to target domain, scale, population, or use? → local validation or block/narrow.

Construct validity belongs in the theory, not an appendix. Validation requires a coding manual, componentwise inter-rater reliability, known-groups and counterexample tests, and discriminant validity from confidence, uncertainty, and severity. It also requires incremental utility over simpler provenance or missing-data flags, calibration of use-specific thresholds, and sensitivity analysis to representation choices (Messick, 1995; Krippendorff, 2018). The vector should be replaced by fewer fields whenever a simpler schema preserves every repair path, blocking set, discharge effect, and gate outcome.

Debt propagates only through recorded use-relevant dependencies and under a declared policy P_d :

$$\mathbf{d}_{\text{out}} = \text{Discharge}_\Delta \left(\text{Join}_{P_d} (\{\delta_t\} \cup \{d_i : i \in \text{Rel}_t\}) \right).$$

δ_t is debt introduced locally; Rel_t is the dependency set relevant to the downstream use. Δ is the set of typed discharge events, and Join_{P_d} is the policy-declared operator. Componentwise maximum with an absorbing unassessed symbol ? defines one strict conservative profile, not a universal debt algebra. Alternative joins are admissible when declared, justified, and tested; false blocking and over-taint are direct falsifiers. Cycles preserve debt by default unless a typed transformation establishes irrelevance. Formal limits, debt, and object transfer remain distinct. Formal limits propagate only through preserved query structure. Debt follows recorded use-relevant dependencies under policy; objects move by choice, while warrant must be re-earned at the target horizon. Dependency soundness is the central information risk. The ledger can propagate only along edges it records or discovers. A hidden relevant edge can create false cleanliness even when every visible transition is correct. The primary stress test therefore seeds known but unrecorded dependencies and measures hidden-edge recall, downstream uncertainty, and whether high-authority use remains blocked. Weak dependency coverage cannot be interpreted as low debt. A coverage claim is audited only when independent reconstruction or audit evidence supports it; it is estimated only when a sampling frame and estimator are declared. Otherwise DependencyCoverage remains Unknown, and authority-sensitive use routes conservatively to review. Discharge and waiver are different types. For component j ,

$$\text{Discharge}_j(\mathbf{d}, \text{evidence}, \text{witness}) \rightarrow \mathbf{d}'$$

changes d_j only when the new evidence is admissible for that component, scope, and horizon. A signature cannot pay unrelated source or transformation debt. A waiver never enters the debt update; it is an input to $\text{Gate}_{\{U,C,g\}}$ that records owner, reason, scope, expiry, residual risk, and revocation conditions. Waiver changes permission while debt remains visible.

The reference gate executes in three phases. First it resolves current source, object, model, policy, authority, credential, and validation versions; tests horizon adequacy $\alpha(U)$; and inspects terminal and evidence status. Next it verifies integrity, freshness, and revocation; computes the relevant debt blocking set; tests authority and role separation; and validates waiver scope

and expiry. Finally it binds any consequential action into Cert_U, verifies the certificate immediately before effect, and emits GateOutcome and GateReceipt. Bounded execution then emits Receipt_exec. A waiver cannot convert HorizonInadequate into Warranted, and PermittedWithWaiver must never be rendered as unqualified green.

CONFORMANCE INVARIANT 5 — conditional debt conservation and non-laundering. Assume complete dependency capture for the named use and complete mediation of presentation and action surfaces. Also assume current evidence, debt, and adequacy inputs; tamper-evident receipts; visible revocation; and pre-effect certificate verification. Finally require atomic binding or commit-time revalidation, plus presentation that derives unqualified success only from Warranted. Under these premises, unresolved threshold-level debt cannot disappear, appear as warranted green, or authorise a stale or substituted action unless a typed discharge pays it. This is a conditional control-plane property. Missing edges, stale data, bypasses, TOCTOU races, duplicate dispatch, receipt substitution, or unauthorised policy changes violate the premises rather than the conclusion.

Transfer moves an object, not its warrant. Reuse across taxon, jurisdiction, organisation, scale, model, or input distribution records origin horizon, target horizon, preserved structure, changed assumptions, required local validation, allowed use now, and blocked use until validation. A transfer passport prevents similarity from masquerading as inherited standing. Cross-domain portability is therefore type portability, not evidence transfer: each domain must instantiate its own horizon, authority, validation, and thresholds; illustrative debt codes do not travel by analogy.

8. Frontier comparators, severe tests, and conditions of defeat

Table 1. Comparator and removal map. Panel A identifies an endpoint-matched comparator, the residual interface, and the failure it should discriminate. Panel B gives the removal rule. No row reports empirical superiority. If a comparator does not represent a DebtVector field, that field is not represented/not assessed; it is never d=0 by default.

A. Comparator, residual interface, and discriminated failure

Functional contract	Endpoint-matched comparator	Residual interface	Failure discriminated
Scientific representation	Causal abstraction; abstract interpretation	Typed levels, Q, fibres/side information, receipts, and loss/recovery/use.	Hidden query loss or undeclared side information.
Executable bounded evaluation	Workflow/operational semantics; runtime verification	Effective operators, scheduler, budgets, terminal sum, receipts, and external-call policy.	Hanging or cyclic execution presented as completion.
Local epistemic transduction	Selective prediction; provenance	Reason-typed non-answer, receipted effects, and candidate/certified state.	Refusal reason or certification status silently collapsed.
Observer composition and meta-control	Provenance + assurance + execution tracing	Versioned observer, history, monitoring, threat/authority metadata, escalation, and pre-effect mediation.	Unmonitored export, stale revocation, or authority drift.
Residual-warrant accounting	Missing-evidence flags; risk registers	Repair-specific vector, policy join, typed discharge, and visible waiver.	False cleanliness or waiver rendered as payment.
Permission and action	Certified traces; firewalls; transactional agents	Horizon adequacy, named use, exact action binding, dispatch receipt, and revalidation.	TOCTOU, substitution, stale/replayed certificate, or duplicate effect.

B. Defeat condition and optional framework label

Functional contract	Explicit defeat condition	Framework label
Scientific representation	Remove when a comparator reconstructs the same query-relative interface and downstream consequences more simply.	RLA
Executable bounded evaluation	Remove when ordinary workflow or transition-system semantics exports the same bounded evaluator.	CRC
Local epistemic transduction	Remove when a simpler reject-option or workflow matches diagnosis, traceability, and safety at lower burden.	ECU / ECNN
Observer composition and meta-control	Remove when provenance + assurance + control reconstruct the observer export and action boundary.	ROA / meta-controller
Residual-warrant accounting	Replace when a simpler schema preserves all repair paths, blocking sets, discharge effects, and gate outcomes.	Epistemic Debt
Permission and action	Replace when certified governance supplies equal semantics and safer outcomes with lower burden.	Gate / action

Matched-baseline rule. A comparator is matched only when it covers the endpoint’s relevant Export_B fields under the same data, horizon, risk, and resource conditions. Use provenance plus assurance for reconstruction; add selective prediction for non-answer; add certified trace, firewall, or transactional control for action. If it reconstructs the same export and protections with lower burden, the named layer has no independent standing.

New methods enter through four outcomes. SUBSTITUTE reconstructs an export at lower burden and removes the layer. STRENGTHEN supplies a better local theorem or certificate. INPUT supplies evidence, monitoring, uncertainty, or a candidate object. BASELINE competes empirically. Compositional, distributionally robust, and transport abstractions can strengthen

or substitute scientific representation. Certified traces challenge bounded evaluation and the gate; execution provenance and AgentTrails challenge observer composition. Composable assurance challenges debt propagation and permission; artificial epistemic-agent work challenges governance meta-control. AEGIS targets pre-execution mediation. Agentic transactions

target atomicity and persistent-state guarantees. Pista targets post-hoc-only human oversight (Felekis et al., 2025, 2026; Lorenz & Tull, 2026; Liu Yanglet, Wang, & Capponi, 2026; Wang et al., 2026; Wu et al., 2026; Zhao, 2026; Marchal et al., 2026; Yuan, Su, & Zhao, 2026; Sun, Wang, & Li, 2026; Sabouri et al., 2026). These are substitution threats, not corroborative authority.

Public frontier-governance frameworks from OpenAI, Anthropic, Google DeepMind, and NIST operate mainly at model, capability, deployment, and organisational-risk levels. They address evaluations, capability thresholds, safeguards, risk reports, residual-risk review, incident response, external review, and lifecycle revision (National Institute of Standards and Technology, 2024; OpenAI, 2026; Anthropic, 2026; Google DeepMind, 2026). The present claim-level architecture neither replaces nor validates them. Its narrower candidate contribution is claim-, artefact-, and action-level semantics that could populate their assurance and monitoring processes. If an organisational framework supplies the same typed semantics and prevents unsupported authoritative use with lower burden, the distinct layer should disappear.

LOCAL PROPOSITION 6 — relative layer removal. Let $A \rightarrow L \rightarrow B$ be adjacent components, and let $\text{Export}_B(L)$ be the complete export of layer L for query family Q under resource policy B . Assume a computable replacement $A \rightarrow B$ reconstructs value/terminal, receipt, evidence and debt effects, policy/version, threat/authority metadata, resource cost, and Q -observables within tolerance. If it also preserves every downstream gate outcome, then L is observationally redundant relative to (Q, B) . The result is relative, not an ontological claim that the concept is useless.

8.1 Evaluation protocol — not a result

PROTOCOL, NOT RESULT. No outcome data are reported. The primary comparative hypothesis is:

Evidence lanes. Machine-only Lab tests cover replay, trace removal, failure visibility, mutation/sensitivity, and bounded dispatch. Domain-expert/data tests cover biological adequacy, calibration, and observable-family discrimination. Human-workflow tests cover reviewer burden, queues, overrides, false-block/unsafe-pass, and decision time. **Matched baselines.** Use provenance + assurance + selective prediction for evidence/reconstruction endpoints, and certified traces + pre-execution firewall/transactional controls for action endpoints. Evidence from one lane cannot upgrade another.

H1. Against the endpoint-matched baseline bundle, the integrated lifecycle reduces unsupported authoritative use and source-path reconstruction time/error without exceeding preregistered limits on false blocking, reviewer minutes, or operational cost.

Core endpoints are false-green rate, unsupported-authoritative-use rate, reconstruction accuracy/time, hidden-edge recall, abstention-reason accuracy, contradiction recall, false-block rate, action-substitution refusal, stale-certificate rejection, duplicate-dispatch rate, gate latency, reviewer burden, override frequency, and time to discharge. Results must be stratified by use risk, domain, subgroup, horizon version, action reversibility, and dependency coverage. Burden proportionality is part of H1: low-risk exploration should not inherit the review cost of high-authority action.

Cost-sensitive support. Let $NB = \Delta\text{Harm} - \sum C_i$, where C_i covers reconstruction, governance, latency, staffing, privacy, and bypass costs. The lifecycle is not operationally supported when $NB \leq 0$, componentwise debt coding fails preregistered reliability after bounded training, or an endpoint-matched baseline dominates the coverage-risk-cost frontier. Reviewer-capacity reporting should include arrival rate, service time, queue length, review latency, bypass, false-block, unsafe-pass, and total human minutes per governed artefact. Safety purchased by unstable queues or systematic bypass is not a win.

Challenge families are grouped by endpoint. Evidence and horizon cases include the same output under different horizons, high confidence without source, broken traces, revoked evidence, hidden dependencies, and bypassed exports. Representation and status cases include hidden side information, unresolved cycles, contradiction coerced into Answer, waiver shown as warrant, unauthorised review, distribution shift, and transferred parameters. Action cases include hanging tools, approve-then-mutate, TOCTOU, stale or substituted certificates, duplicate or partial execution, forbidden subagent composition, receipt substitution, stop/effect races, and post-state invalidation. Search-augmented systems require explicit abstention tests because added retrieval can improve answerable-query accuracy while worsening behaviour on unanswerable queries (Xie et al., 2026). Certified traces, pre-execution firewalls, execution provenance, transactional agents, and interactive oversight supply severe agentic baselines (Liu Yanglet, Wang, & Capponi, 2026; Yuan, Su, & Zhao, 2026; Wang et al., 2026; Sun, Wang, & Li, 2026; Sabouri et al., 2026).

The programme-level defeat condition is explicit. Independent preregistered studies may show that endpoint-matched simpler systems weakly dominate on unsupported use, safe action control, reconstruction, and burden. In that case the architecture must contract to those systems or become a domain-specific design pattern. Internal consistency, implementation polish, or self-application cannot waive this loss rule.

8.2 Limitations and open debt

These limits are already load-bearing in the preceding contracts; this subsection consolidates rather than adds them.

- Dependency coverage. Debt propagates only through recorded or discovered use-relevant dependencies; the architecture does not guarantee complete hidden-edge discovery (§7).
- Horizon construction. Declaring $\mathcal{H}$ enables local computation, but selecting an adequate horizon for U is a governance obligation and can be wrong or self-serving (§2).
- Instrument reliability. The nine-component DebtVector remains a candidate ordinal/categorical instrument pending coding-manual tests, inter-rater reliability, construct/discriminant validity, and incremental utility (§7).
- Human and latency burden. Review minutes, gate latency, false blocking, queue instability, and bypass are comparative costs, not automatic safety gains (§8.1).
- Empirical superiority and transfer. Constructibility and formal coherence do not establish comparative benefit or cross-domain generalisation; heterogeneous preregistered studies remain required (§8.1).

9. Discussion and conclusion

The centre of gravity is a bounded computational semantics of claim admissibility and use-readiness. Ordinary effective evaluation returns evidence status, reason-typed non-answer states, Epistemic Debt, and a reconstructable receipt. A distinct gate computes permission; consequential effects require a current action certificate and execution receipt. The functional contracts are scientific representation, bounded evaluation, local epistemic transduction, persistent observer composition, governance meta-control, residual-warrant accounting, and action mediation. Framework labels remain secondary and disappear when the same exports are reconstructed more simply.

The architecture is intentionally defeasible. Causal abstraction can subsume scientific representation; ordinary workflow semantics can subsume bounded evaluation; selective prediction can subsume the epistemic head. Certified traces, execution provenance, pre-execution firewalls, transactional agents, and composable assurance can subsume observer or action control; a simpler missing-evidence schema can replace the debt vector. The architecture earns its complexity only when interfaces

among these methods expose failures that isolated components miss or materially improve reconstruction, unsupported-use control, and safe action.

Nothing in this tightening discharges empirical debt. Coding anchors are protocol, not reliability evidence. The worked path is an executable specification, not a superiority result, and DependencyCoverage remains unproven. The tests in §8.1 and the limitations below state what must still be shown.

The result is not a universal truth machine, nor a claim that meaning has been reduced to computation. It is a bounded semantics of admissibility, use-readiness, and governed action that respects formal limits. Its strongest virtue is honest termination and interposition. Answer, unknown, contradiction, out-of-horizon, review, timeout, failure, or blocked effect each carry a receipt and a condition for state change. When semantic artefacts circulate faster than institutions can judge them, six questions matter: what has the artefact earned; what does it still owe; which path supports it; which action is certified; what happened; and who authorised it?

Declarations and reproducibility

The author is responsible for the framework, formal choices, interpretations, and remaining errors. AI-assisted tools supported restructuring, consistency checking, code generation, and language editing; no AI system is treated as scientific authority. No empirical outcome study is reported. A conforming implementation should publish a versioned schema, reference evaluator, frozen conformance corpus, positive and negative tests, threat model, and monitored-use/action-surface inventory. It should also publish receipt readback, policy/horizon identifiers, a certificate verifier, and bounded external-call adapters. Benchmark code should report epistemic benefit, false blocking, reviewer burden, gate latency, action-substitution refusal, cost, and disparate effects; idempotency or compensation rules are required where needed.

Repository and executable companion. The expanded framework and its source corpus are available at the full repository URL below. The browser laboratory is an implementation witness and expert-facing falsification surface; it is not biological validation.

Repository: <https://github.com/Luke883i/ROA>

Bryophyte Epistemic Lab v9 (standalone HTML source/download page):

https://github.com/Luke883i/ROA/blob/main/Reticular%20Local%20Abstraction%20RLA-CRC-ECNN/ROA%20Bryophyte%20Epistemic%20Lab/ROA_Bryophyte_Epistemic_Lab_v9.html

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